

Sprint Retrospective Report

Campus Food Ordering Platform

Consolidated summary of retrospective meetings across four sprints

Summary

Across the four sprints, our team moved from project setup and basic feature scaffolding to a more complete platform covering authentication, menu management, order tracking, online payments, SA dietary and allergen data integration, analytics, and system stabilisation. Ledile focused on menu management and product data, Faranani coordinated the team as Scrum Master and led authentication work, Tshepo built the payment and tracking foundations, Thato supported the payment system and food recommendation work, Olwethu handled analytics across the project, and Senzeni contributed to payment and SA data integration tasks. The retrospectives show steady growth in technical depth, collaboration, testing, and architectural awareness. Early sprints focused on getting the core structure working, while later sprints concentrated on integration, security, maintainability, and polishing the final product.

Sprint 1 - Foundation and project setup

What went well: Ledile built the menu-management modules across the frontend and backend, Faranani set up the repository, GitHub Projects, Firebase, and the authentication foundation, and Tshepo scaffolded the initial payment components. This gave the team a practical starting point and helped connect the main parts of the platform early.

What could be improved: The sprint would have benefited from earlier alignment on task scope, folder structure, and role permissions. Duplicate filenames between frontend and backend created confusion, and some time was lost clarifying requirements before implementation.

Action items for next sprint: Agree on file naming and folder boundaries before coding begins, confirm task understanding with teammates and the Scrum Master, document the architecture and role model early, and split larger tasks into smaller, testable pieces to reduce rework.

Sprint 2 - Refinement, integration, and stability

What went well: Faranani helped clarify the three-role model and coordinate integration, Ledile moved menu data toward Firestore, Tshepo advanced the order-tracking foundation, and Thato supported the payment work. The system became more stable and better aligned with the project requirements.

What could be improved: Merge conflicts, inconsistent data structures, and late verification of end-to-end flows slowed progress at times. A shared schema would have reduced debugging around customer IDs and order records.

Action items for next sprint: Document the expected data fields for shared collections and APIs, run integration checks before merging each feature branch, ask team members to demonstrate working features earlier, and maintain one shared checklist for authentication, ordering, and tracking dependencies.

Sprint 3 - Feature completion and security

What went well: Tshepo led the payment workflow using Paystack, Faranani supported integration and security rules, Ledile improved the product-management interface and standardised dietary and allergen information, and Senzeni contributed to the payment system and SA data integration. The platform became more complete and practical for students.

What could be improved: The payment flow and product logic were complex enough that more upfront design documentation would have helped. Some UI and validation logic could also be separated more clearly from business logic to make testing easier.

Action items for next sprint: Document the payment architecture and data flow before adding more features, keep business logic, UI logic, and data access in separate modules where possible, add regression tests for payment, menu validation, and order-status changes, and share short implementation notes when features affect more than one team member's code.

Sprint 4 - Stabilisation, testing, and final polish

What went well: Faranani helped stabilise the project and improve test coverage, Ledile contributed to analytics design and architectural planning, Tshepo supported final integration and testing, Olwethu continued the analytics work, and Thato contributed the food recommendation features. The team successfully brought the major flows together into a more dependable application.

What could be improved: Leaving too much work until the end created stress, some tasks relied too heavily on a few members, and a few completed items still needed verification before integration.

Action items for next sprint: Track progress more frequently, spread ownership more evenly so the team can keep moving when one member is busy, increase test coverage before final integration, and begin stabilisation earlier so there is enough time for review and refinement.

Alignment with Project Requirements

The retrospectives show that the team progressively addressed the main project requirements: user verification, menu management, order management, order tracking, payments, SA data integration, and analytics. Faranani and the authentication work supported the 3-role verification requirement, Ledile and the vendor features covered menu management, Tshepo and the tracking work supported order management and real-time updates, the payment work met the online payment requirement, and the SA data and analytics features were carried through by multiple team members across the project.

Requirement area	Evidence from the retrospective summaries
User verification	Third-party authentication, role-based access, and admin application flows were established and refined.
Menu and order management	Vendor menu tools, product editing, sold-out handling, and order-processing workflows were built and improved.
Order tracking	Real-time tracking with REST and Socket.io support was introduced and then stabilised.
Payments	A secure online payment workflow was implemented, verified, and integrated into the order lifecycle.
SA data integration	Dietary and allergen information was standardised using a public food data source and predefined selections.
Analytics	Analytics design and reporting work advanced alongside the final integration and polish stage.

Conclusion

The project matured from an initial working structure into a more complete and dependable system. Across the four sprints, Ledile, Faranani, Tshepo, Thato, Olwethu, and Senzeni each contributed to building a stronger final product and a better team workflow. The main lessons were the value of planning, clear communication, early testing, and well-defined architecture. By the final sprint, the team had improved both the product and the way it was built.